Data Analyst

Building a Strong Foundation

- Introduction
  - What is Data Analytics
  - Types of Data Analytics
    - Descriptive Analytics
    - Diagnostic Analytics
    - Predictive Analytics
    - Prescriptive Analytics
- Key Concepts of Data
  - Collection
  - Cleanup
  - Exploration
  - Visualisation
  - Statistical Analysis
  - Machine Learning
- Analysis / Reporting with Excel
  - IF
  - DATEDIF
  - VLOOKUP / HLOOKUP
  - REPLACE / SUBSTITUTE
  - UPPER / LOWER / PROPER
  - CONCAT
  - TRIM
  - AVERAGE
  - COUNT
  - SUM
  - MIN / MAX
  - Learn Common Functions
- Charting
- Pivot Tables
- Learn SQL

Gain Programming Skills

1. Learn a Programming Lang. (Python, R)
2. Data Manipulation Libraries — Pandas, Dplyr
3. Data Visualisation Libraries — Matplotlib, Ggplot2

Mastering Data Handling

- Data Collection
  - Databases
  - CSV Files
  - APIs
  - Web Scraping
- Data Cleanup
  - Handling Missing Data
  - Removing Duplicates
  - Finding Outliers
  - Data Transformation
  - Using Libraries for Cleanup: Pandas, Dplyr

Data Analysis Techniques

- Descriptive Analysis
  - Generating Statistics
    - Dispersion
      - Range
      - Variance
      - Standard Deviation
    - Central Tendency
      - Mean
      - Median
      - Mode
      - Average
    - Distribution Space
      - Skewness
      - Kurtosis
  - Visualizing Distributions
- Statistical Analysis
  - Learn to analyze relationships and make data driven decisions
  - Hypothesis Testing
  - Correlation Analysis
  - Regression
  - Learn Different Techniques
- Data Visualisation
  - Tools
    - Tableau
    - Power BI
  - Libraries
    - Matplotlib
    - ggplot2
    - Seaborn
  - Charting
    - Bar Charts
    - Histograms
    - Line Chart
    - Stacked Charts
    - Scatter Plot
    - Heatmap
    - Funnel Charts
    - Pie Charts

Advanced Topics

- Machine Learning
  - Machine Learning Types
    - Reinforcement Learning
    - Unsupervised Learning
    - Supervised Learning
  - Popular ML Algorithms
    - Decision Trees
    - Naive Byes
    - KNN
    - K-Means Clustering
    - Logistic Regression
  - Model Evaluation Techniques
- Deep Learning (Optional)
  - Frameworks
    - Tensorflow
    - Pytorch
  - CNNs
  - RNN
  - Neural Networks
  - Learn the Basics
  - Practice Training Models
  - Image Recognition
  - Natural Language Processing
- Big Data Technologies
  - Big Data Concepts
  - Data Storage Solutions
  - Data Processing Frameworks
    - Hadoop
    - Spark
  - Data Processing Techniques
    - Parallel Processing
    - MPI
    - MapReduce

Build a portfolio of projects. Some example projects:

1) Predicting sales trends 2) Customer Segmentation.

Participate in Kaggle Competitions

Participate in competitions and learn from community

Online Courses and Certifications

Platforms like Coursera, edX, Udemy and Datacamp offer courses in Data Analytics and related topics.

Stay Updated and Network

Join networking events, communities, seek mentorship, attend webinars and engage in discussions to expand your network.